



Title: Lab Project

Course Code : CSE282.4

Course Title : **Programming Language II Lab**

Department : CSE

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Work Place : **SOUTH EAST UNIVERSITY**

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Lost & Found Management System (LFMS).

1. INTRODUCTION:

Project Overview:

The Lost & Found Management System (LFMS) is a Java-based application designed to assist users in reporting lost items, registering found items, and submitting ownership claims. The system provides separate functionalities for regular users and administrators to ensure secure and organized management of lost-and-found records.

Project Objectives:

1. Provide a platform for reporting lost items.
2. Allow users to register found items.
3. Enable ownership claim submission and verification.
4. Maintain secure user authentication and authorization.
5. Apply Object-Oriented Programming (OOP) principles in software development.

2. SOFTWARE DEVELOPMENT PROCESS:

The project requirements were analyzed and divided into the following modules:

User Module:

1. Registration
2. Login
3. Profile management

Item Management Module:

1. Add lost item
2. Add found item
3. View items

Claim Management Module:

1. Submit claim
2. Review claim status

Admin Module:

1. Approve claims
2. Reject claims
3. Manage records

3. DEVELOPMENT METHODOLOGY:

An incremental development approach was followed.

Phase 1

- Model Classes

Phase 2

- Service Classes

Phase 3

- Authentication System

Phase 4

- Claim Management

Phase 5

- User Interface Integration(GUI)

Phase 6

- Testing and Bug Fixing

4. SYSTEM DESIGN:

Package Structure:

```
src
├── model
├── service
├── exception
├── gui
└── main
```

1.Model Package

Stores system entities.

2.Service Package

Contains business logic.

3.Exception Package

Handles custom exceptions.

4.GUI Package

Contains graphical user interface components.

5.Main Package

Application entry point.

5.SYSTEM ARCHITECTURE:

The application follows a layered architecture:

User Interface Layer



Service Layer



Model Layer

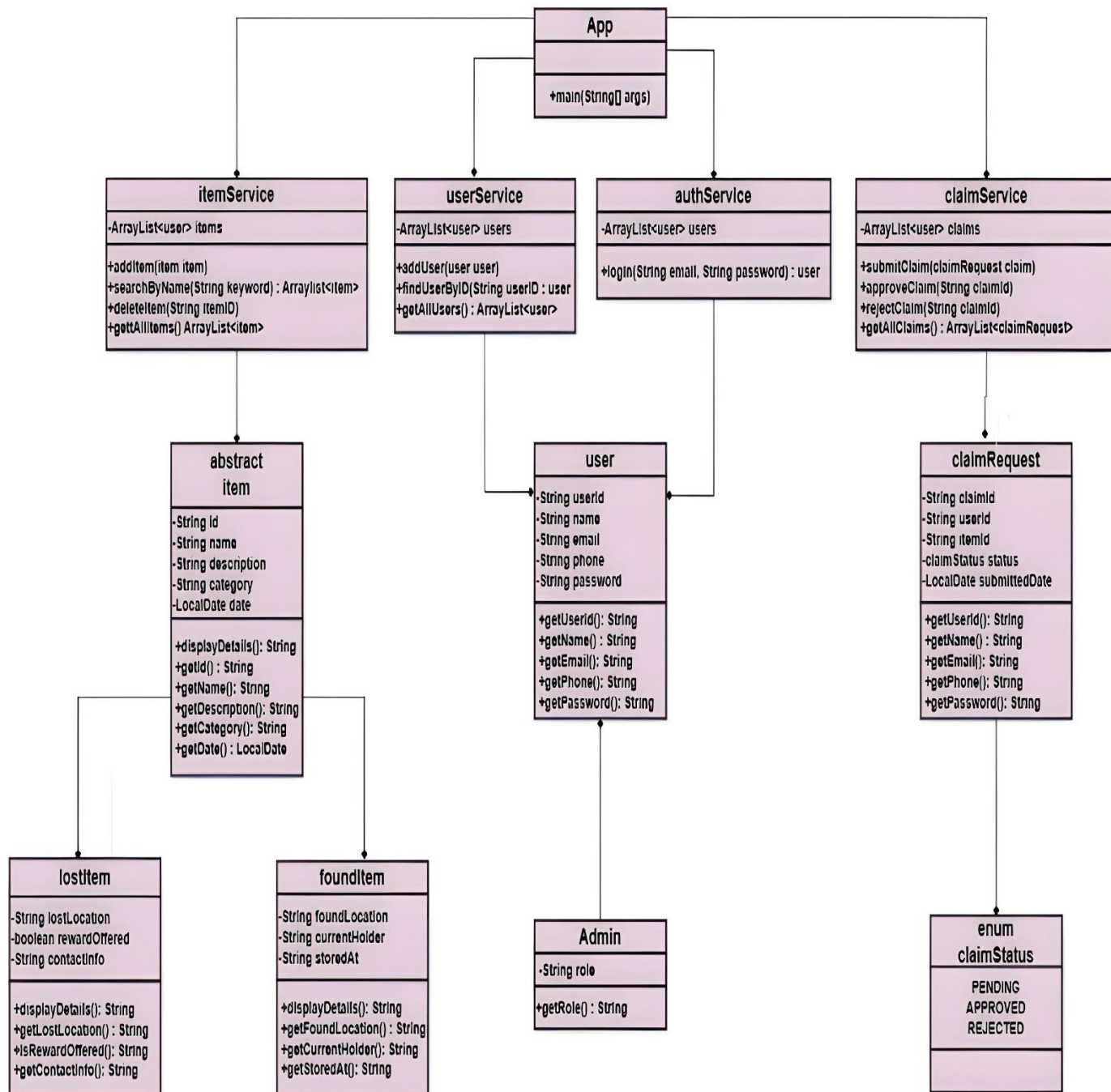


File Storage Layer

6.CLASS DESIGN

CLASS	DESCRIPTION
App	The main driver function that run all the other classes and methods .
Item (abstract)	Base class defining shared item fields.
LostItem	Extends Item with lostLocation, rewardOffered, contactInfo.
FoundItem	Extends Item with foundLocation, storedAt, currentHolder.
User	Holds personal details and item-related actions.
Admin	Inherits from User with additional privileges (approve/reject claims).
ClaimStatus	To verify if the item has been claimed, Approved or Pending
ClaimRequest	Links users and items through claim processing and validation.
authService	To make sure that users don't accidentally get admin access .
ClaimService	Making sure that the claim process is well documented and valid via exception .
itemService	Ensuring that the item input is valid and no duplicates are made via exception .
userService	Ensuring that the user input is valid and no duplicates are made via exception .

7. CLASS DIAGRAM:



8. OOP Concept Mapping:

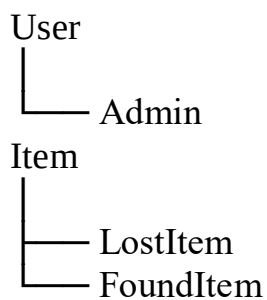
Encapsulation:

Encapsulation was achieved by declaring class attributes as private and providing controlled access through getter and setter methods.

Classes demonstrating encapsulation:

- ❖ User
- ❖ Admin
- ❖ Item
- ❖ LostItem
- ❖ FoundItem
- ❖ ClaimRequest

Inheritance:



Abstraction:

The Item class was designed as an abstract class that defines common item properties while allowing subclasses to implement specific behavior.

Abstract Class:

- ❖ Item

Concrete Implementations:

- ❖ LostItem
- ❖ FoundItem

Polymorphism:

Polymorphism was implemented through method overriding.

Examples:

- LostItem.displayDetails()
- FoundItem.displayDetails()

Objects of different subclasses can be treated as Item objects while executing their own specialized implementations.

Exception Handling:

Custom exception handling was integrated into service classes to ensure data integrity and system reliability.

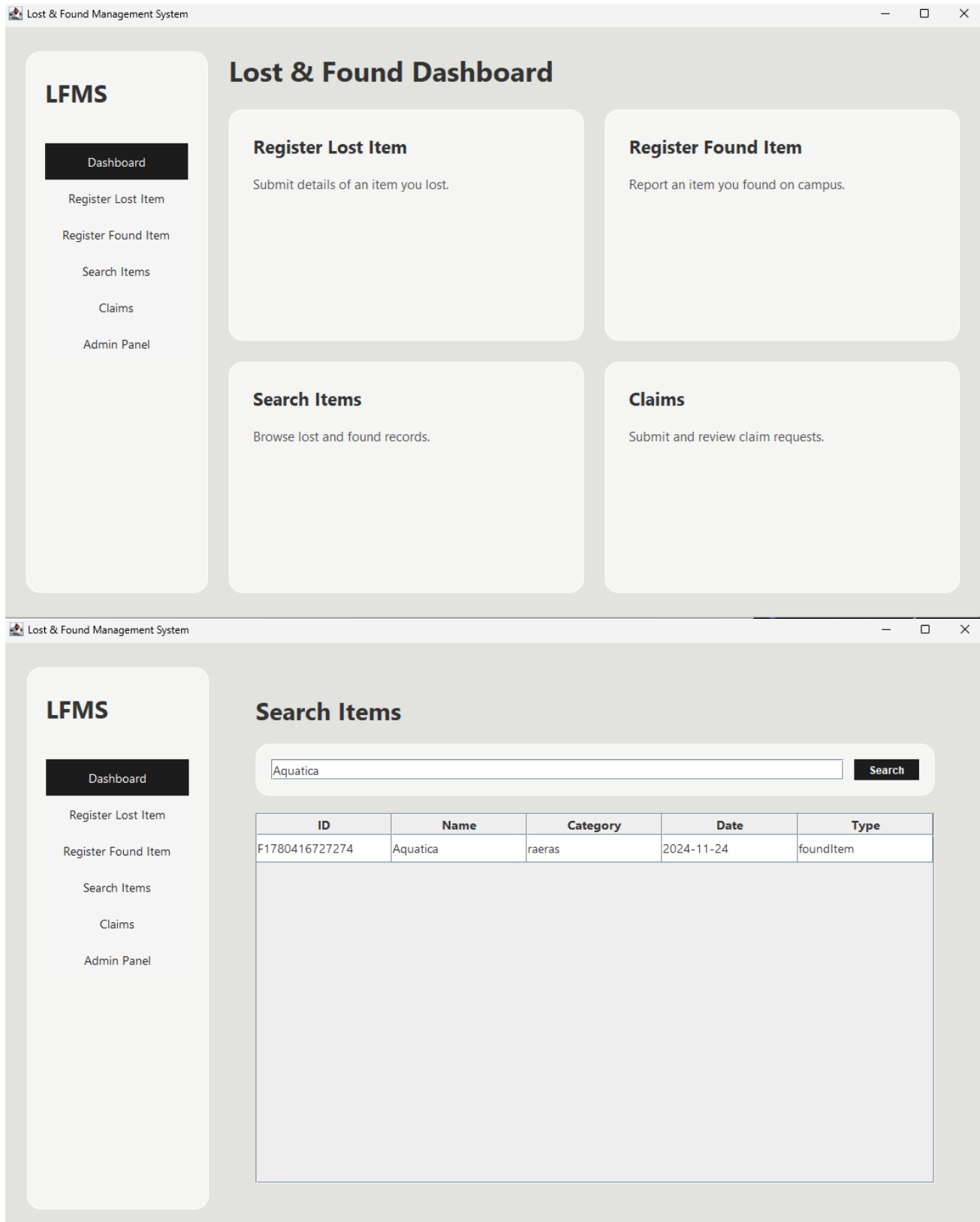
Service Classes Using Exceptions:

- UserService
- ItemService
- ClaimService
- AuthService

Examples:

- DuplicateUserException
- DuplicateItemException
- UserNotFoundException
- ItemNotFoundException
- ClaimNotFoundException

9.GUI SCREENSHOTS:



Lost & Found Management System

Lost & Found Dashboard

LFMS

- Dashboard
- Register Lost Item
- Register Found Item
- Search Items
- Claims
- Admin Panel

Register Lost Item
Submit details of an item you lost.

Register Found Item
Report an item you found on campus.

Search Items
Browse lost and found records.

Claims
Submit and review claim requests.

Lost & Found Management System

Search Items

Aquatica **Search**

ID	Name	Category	Date	Type
F1780416727274	Aquatica	raeras	2024-11-24	foundItem

The screenshot displays the 'Lost & Found Management System' (LFMS) web application. The interface is divided into a left sidebar and a main content area. The sidebar, titled 'LFMS', contains a 'Dashboard' button (highlighted in black) and several menu items: 'Register Lost Item', 'Register Found Item', 'Search Items', 'Claims', and 'Admin Panel'. The main content area is titled 'Register Lost Item' and includes the instruction 'Submit details of an item you lost.' Below this, there are several input fields: 'Item Name', 'Description', 'Category', 'Date Lost (YYYY-MM-DD)', 'Lost Location', and 'Contact Information'. A checkbox labeled 'Reward Offered' is located at the bottom of the form. The browser window title is 'Lost & Found Management System'.

LFMS

Dashboard

Register Lost Item

Register Found Item

Search Items

Claims

Admin Panel

Register Lost Item

Submit details of an item you lost.

Item Name

Description

Category

Date Lost (YYYY-MM-DD)

Lost Location

Contact Information

☐ Reward Offered

The screenshot displays the 'Lost & Found Management System' (LFMS) web application. The interface is divided into a left sidebar and a main content area. The sidebar, titled 'LFMS', contains a 'Dashboard' button (highlighted in black) and several menu items: 'Register Lost Item', 'Register Found Item', 'Search Items', 'Claims', and 'Admin Panel'. The main content area is titled 'Register Found Item' and includes the instruction 'Submit details of a found item.' Below this, there are several input fields: 'Item Name', 'Description', 'Category', 'Date Found (YYYY-MM-DD)', 'Found Location', and 'Current Holder'. The browser window title is 'Lost & Found Management System'.

LFMS

Dashboard

Register Lost Item

Register Found Item

Search Items

Claims

Admin Panel

Register Found Item

Submit details of a found item.

Item Name

Description

Category

Date Found (YYYY-MM-DD)

Found Location

Current Holder

Lost & Found Management System

LFMS

Dashboard

Register Lost Item

Register Found Item

Search Items

Claims

Admin Panel

Submit Claim Request

Claim an item that you believe belongs to you.

User ID

11234

Item ID

L1780417171617

Submit Claim

Message

Claim submitted successfully!

OK

Lost & Found Management System

LFMS

Dashboard

Register Lost Item

Register Found Item

Search Items

Claims

Admin Panel

Admin Panel

Items

Claims

ID	Type	Name	Category	Date
F1780416727274	foundItem	Aqualica	raeras	2024-11-24
L1780417171617	lostItem	Marwa	faefsa	2024-11-12
L1780418653174	lostItem	Lotion	Cosmetics	2026-11-22
L1780419620327	lostItem	yay	yay	2026-05-05

Refresh ItemsDelete Selected Item

Lost & Found Management System

LFMS

Dashboard

Register Lost Item

Register Found Item

Search Items

Claims

Admin Panel

Item Name

Description

Category

Date Found (YYYY)

Found Location

Current Holder

Stored At

Submit Found Item

Enter Admin Password

OK

Cancel

Lost & Found Management System

LFMS

Dashboard

Register Lost Item

Register Found Item

Search Items

Claims

Admin Panel

Admin Panel

Items

Claims

Claim ID	User ID	Item ID	Status	Submitted Date
C1780418151020	F1001	F1780416727274	APPROVED	2026-06-02
C1780418689664	Rayhan01	L1780417171617	PENDING	2026-06-02
C1780419675712	123456	L1780419620327	APPROVED	2026-06-02
C1780419764321	123456	L1780419620327	APPROVED	2026-06-02

Refresh Claims

Approve Claim

Reject Claim

9. REFLECTION ON CHALLENGES FACED:

This project provided practical experience in applying Object-Oriented Programming concepts to a real-world problem. Through collaborative development, the team gained hands-on experience with Git version control, branch management, and software integration.

The project reinforced the importance of proper system design, modular programming, and teamwork. Challenges such as merge conflicts, access control implementation, exception handling, and GUI integration improved our problem-solving abilities and understanding of software engineering practices. The GUI development caused several issues during implementation, especially in terms of layout management and event handling. In addition, designing and implementing the admin panel was a particularly complex task due to the need for strict access control and multiple administrative functions.

Overall, the Lost & Found Management System successfully demonstrated the use of Java, OOP principles, file handling, exception management, GUI development, and collaborative software development techniques.

10. Conclusion:

The Lost & Found Management System was successfully designed and implemented using Java and Object-Oriented Programming principles. The system fulfills its primary objectives of managing lost items, found items, user authentication, and ownership claims. The project also provided valuable experience in software design, implementation, testing, and team-based development using Git and GitHub.